MTH 200 Chapter 3: Derivatives

3.9 Derivatives of Exponential and Logarithmic Functions

Recall: In section 3.6 we established the following theorem.

> **Theorem:** Suppose that b is a positive real number constant. Then,
> $$\frac{d}{dx}\left(b^x\right) = b^x \ln(b)$$

Note: This derivative formula agreed with the fact that $\frac{d}{dx}\left(e^x\right) = e^x$.

Example 1: Find the first derivative of $y = 3^{\tan(x)}$.

We note that $y' = 3^{\tan(x)} \ln(3) \cdot \frac{d}{dx}\left[\tan(x)\right] = 3^{\tan(x)} \ln(3) \cdot \sec^2(x)$

Since we have already seen how to take the derivative of exponential functions we turn our attention to logarithmic functions.

Motivation: We would like to find a formula for the derivative of $y = \log_b(x)$ where $b > 0$ and $x > 0$.

Our strategy will be to do this with implicit differentiation.
By definition recall that $y = \log_b(x)$ means that $b^y = x$. Therefore we compute:

$$b^y = x \quad \Rightarrow \quad \frac{d}{dx}\left(b^y\right) = \frac{d}{dx}(x) \quad \Rightarrow \quad b^y \ln(b)\frac{dy}{dx} = 1 \quad \Leftrightarrow \quad \frac{dy}{dx} = \frac{1}{b^y \ln(b)} = \frac{1}{x\ln(b)}$$

Question: What does this fact imply about the first derivative of $\ln(x)$*?*

> **Theorem:** Suppose that b is a positive real number constant. Then,
> $$\frac{d}{dx}\left[\log_b(x)\right] = \frac{1}{x\ln(b)}$$

Note: Again this implies that $\frac{d}{dx}\left[\ln(x)\right] = \frac{1}{x}$.

Example 2: Answer each of the following.

a) Find the first derivative of $y = \ln(\sin(x))$.

We note that $y' = \dfrac{1}{\sin(x)} \cdot \dfrac{d}{dx}\left[\sin(x)\right] = \dfrac{1}{\sin(x)} \cdot \cos(x) = \dfrac{\cos(x)}{\sin(x)} = \cot(x)$.

b) Find the first derivative of $y = \log_{10}(2x + \cos(x))$.

We note that $y' = \dfrac{1}{(2x+\cos(x))\ln(10)} \cdot \dfrac{d}{dx}\left[2x + \cos(x)\right] = \dfrac{2 - \sin(x)}{(2x+\cos(x))\ln(10)}$.

c) Find the first derivative of $y = \ln|x|$.

We note that $y = \begin{cases} \ln(x) & \text{if } x > 0 \\ \ln(-x) & \text{if } x < 0 \end{cases}$. So, $\dfrac{dy}{dx} = \begin{cases} \dfrac{1}{x} & \text{if } x > 0 \\ \dfrac{1}{-x} \cdot (-1) & \text{if } x < 0 \end{cases} = \dfrac{dy}{dx} = \begin{cases} \dfrac{1}{x} & \text{if } x > 0 \\ \dfrac{1}{x} & \text{if } x < 0 \end{cases} = \dfrac{1}{x}$

Definition: Logarithmic Differentiation is a process of differentiating a function by first taking the logarithm of both sides and then using implicit differentiation to find the derivative.

Note: Properties of logarithms will allow us to make complicated functions much easier to differentiate.

Properties of Logarithms:

Product Rule: $\log_a(xy) = \log_a(x) + \log_a(y)$

Quotient Rule: $\log_a\left(\dfrac{x}{y}\right) = \log_a(x) - \log_a(y)$

Power Rule: $\log_a(x^n) = n \cdot \log_a(x)$

Note: The necessary restrictions on x, y, a, & n have been omitted here.

Example 3: Use logarithmic differentiation to find the first derivative of $y = \dfrac{\sqrt{x^2+1}}{(3x+2)^5}$.

We start by taking the natural logarithm of both sides of this equation.

$$y = \frac{\sqrt{x^2+1}}{(3x+2)^5} \quad \Leftrightarrow \quad \ln(y) = \ln\left(\frac{\sqrt{x^2+1}}{(3x+2)^5}\right)$$

Using properties of logarithms we simplify the RHS and obtain:

$$\ln(y) = \ln\left(\frac{\sqrt{x^2+1}}{(3x+2)^5}\right) \quad \Leftrightarrow \quad \ln(y) = \ln\left(\sqrt{x^2+1}\right) - \ln\left((3x+2)^5\right) \quad \Leftrightarrow \quad \ln(y) = \frac{1}{2}\ln\left(x^2+1\right) - 5\ln(3x+2)$$

Then by taking the derivative of both sides of the equation we obtain:

$$\frac{d}{dx}\left[\ln(y)\right] = \frac{d}{dx}\left[\frac{1}{2}\ln\left(x^2+1\right) - 5\ln(3x+2)\right]$$

$$\Leftrightarrow \frac{1}{y}\frac{dy}{dx} = \frac{1}{2}\cdot\frac{1}{x^2+1}\cdot 2x - 5\cdot\frac{1}{3x+2}\cdot 3$$

$$\Leftrightarrow \frac{dy}{dx} = y\left[\frac{x}{x^2+1} - \frac{15}{3x+2}\right]$$

$$\Leftrightarrow \frac{dy}{dx} = \frac{\sqrt{x^2+1}}{(3x+2)^5}\left[\frac{x}{x^2+1} - \frac{15}{3x+2}\right]$$

Note: This problem could have also been done by using the quotient rule and chain rule, however the process may have been slightly more difficult.

In the above example the process of logarithmic differentiation was not *needed*. The next example shows a time when this process is imperative.

Example 4: Find the first derivative of $y = x^x$, assuming $x > 0$.

None of the rules for differentiation we have learned thus far allow us to immediately find $\dfrac{dy}{dx}$. So, we apply logarithmic differentiation.

$$y = x^x \quad \Rightarrow \quad \ln(y) = \ln\left(x^x\right) \quad \Leftrightarrow \quad \ln(y) = x\cdot\ln(x)$$

$$\Leftrightarrow \quad \frac{d}{dx}\left[\ln(y)\right] = \frac{d}{dx}\left[x\cdot\ln(x)\right]$$

$$\Leftrightarrow \frac{1}{y}\frac{dy}{dx} = 1\cdot\ln(x) + x\cdot\frac{1}{x}$$

$$\Leftrightarrow \frac{dy}{dx} = y\left[\ln(x)+1\right]$$

$$\Leftrightarrow \frac{dy}{dx} = x^x\left[\ln(x)+1\right]$$

Note: The selection to take the <u>natural logarithm</u> of both sides of the equation in the previous examples is arbitrary. We could always choose to complete this process using a logarithm of any base, however since the natural logarithm has a much simpler derivative formula we often choose it.

<u>Example 5:</u> Use logarithmic differentiation to prove the power rule.

Suppose that $y = x^n$ where n is an arbitrary constant, and x only takes on real number values which make y a real number. We note that: $y = x^n \quad \Rightarrow \quad |y| = |x|^n \quad \Rightarrow \quad \ln|y| = \ln\left(|x|^n\right) \quad \Rightarrow \quad \ln|y| = n\ln|x|$.

Therefore we compute:

$$\begin{aligned} \ln|y| = n\ln|x| \quad &\Rightarrow \quad \frac{d}{dx}\left(\ln|y|\right) = \frac{d}{dx}\left(n\ln|x|\right) \\ &\Rightarrow \quad \frac{1}{y}\cdot\frac{dy}{dx} = n\cdot\frac{1}{x} \\ &\Rightarrow \quad \frac{dy}{dx} = \frac{ny}{x} = \frac{nx^n}{x} = nx^{n-1}. \end{aligned}$$

This proves the desired result.

■

The following is an interesting fact which is oftentimes quite useful.

Fact: $\lim\limits_{x\to 0}(1+x)^{\frac{1}{x}} = \lim\limits_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$

Explanation (if time permits):

If $f(x) = \ln(x)$, then $f'(x) = \frac{1}{x}$.

Moreover, $1 = f'(1) = \lim\limits_{h\to 0}\frac{f(1+h)-f(1)}{h} = \lim\limits_{h\to 0}\frac{\ln(1+h)-\ln(1)}{h} = \lim\limits_{h\to 0}\left[\frac{1}{h}\ln(1+h)\right] = \lim\limits_{h\to 0}\left[\ln\left((1+h)^{\frac{1}{h}}\right)\right]$.

Then, $\lim\limits_{h\to 0}\left[\ln\left((1+h)^{\frac{1}{h}}\right)\right] = 1 \Rightarrow e^{\lim\limits_{h\to 0}\left[\ln\left((1+h)^{\frac{1}{h}}\right)\right]} = e \Rightarrow \lim\limits_{h\to 0} e^{\ln\left((1+h)^{\frac{1}{h}}\right)} = e \Rightarrow \lim\limits_{h\to 0}(1+h)^{\frac{1}{h}} = e.$